

GigAI

AI-Powered Construction Intelligence Platform

Comprehensive Technical Report

Version 2.0 | March 21, 2026

CONFIDENTIAL

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1. Executive Summary

GigAI is an AI-powered construction project management platform that automates material change coordination, project communication intelligence, schedule prediction, and cross-stakeholder notification for construction project managers. The system reduces PM coordination time by 80% (from 4+ hours to under 12 minutes per material change) by automating the capture, enrichment, analysis, and execution of coordination workflows.

1.1 Problem Statement

Construction project managers spend 60-70% of their coordination time on manual tasks: transcribing meeting notes, contacting suppliers for quotes, updating schedules and drawings, notifying stakeholders across channels, and following up on procurement. Academic research (PMI, McKinsey, Deloitte) shows that 29% of construction projects fail due to poor communication, 26% of all rework is caused by miscommunication, and PMs spend 45% of their time on administrative overhead. The construction industry is the second-least digitized sector globally, with labor productivity growing at only 1% per year over two decades.

1.2 Solution Architecture

GigAI implements an event-driven AI pipeline that captures communications from multiple sources (email, meeting transcripts, ACC notifications), processes them through a multi-stage AI enrichment pipeline using Claude LLMs, generates coordinated action proposals with confidence scoring, and executes approved actions automatically. Version 2.0 extends this with a unified inbox, RFI automation, decision tracking, stakeholder communication mapping, change impact simulation (EVA + Monte Carlo), auto-generated reports, and smart notification routing.

1.3 Key Metrics

Metric	Value
Total API Endpoints	50+
Backend Test Coverage	235 tests passing
Frontend Tests	16 tests passing
LLM Models Used	Claude Sonnet 4 + Haiku 4.5
Embedding Model	Voyage-3 (1024 dimensions)
Database	PostgreSQL 15 + pgvector
Projects in Library	30 across 6 categories
Employees in Directory	30 across 6 professions
Inbox Communications	44 across 4 sources
Auto-detected RFIs	17

Decisions Tracked	8
Schedule Tasks	24 with dependencies
ACC Data Categories	16 (budgets, contracts, RFIs, etc.)
Smart Notifications	44 with 3-tier routing

2. System Architecture

2.1 Technology Stack

Layer	Technology	Purpose
Backend	Python 3.11+ / FastAPI	Async-native REST API server
AI - Proposals	Claude Sonnet 4	Complex analysis, proposal generation, RFI drafting
AI - Classification	Claude Haiku 4.5	Fast normalization, routing, classification
Embeddings	Voyage-3 (1024-dim)	Semantic search with asymmetric retrieval
Database	PostgreSQL 15 + pgvector	Structured data + vector similarity search
Event Bus	Google Cloud Pub/Sub	At-least-once event delivery (async path)
External APIs	Autodesk Construction Cloud	16 API categories for project data
Frontend	React 18 + Vite + Tailwind	Real-time dashboard with SSE
Design System	Custom CSS (Apple dark mode)	Inter font, warm grays, system colors

2.2 Directory Structure

```
src/ - Python backend (FastAPI + pipeline)

src/api/ - REST API routes (proposals, projects, employees, schedule, inbox, RFIs, decisions, ACC)

src/inbox/ - Unified inbox (models, store, classifier, seed, routes)

src/rfi/ - RFI automation (models, store, detector, drafter, seed, routes)

src/decisions/ - Decision tracker (models, store, detector, seed, routes)

src/stakeholders/ - Stakeholder communication map (graph, routes)

src/impact/ - Change impact simulator (EVA + Monte Carlo, routes)

src/reports/ - Auto-generated reports (generator, routes)

src/notifications/ - Smart notifications (scoring, routing, routes)

src/intelligence/ - GigAI learning engine (decision patterns, vector storage)

src/pipeline/ - Event processing pipeline runner

src/data/ - Project library (30 projects), Employee library (30 employees), Schedules

src/shared/ - Pydantic models, DB clients, LLM wrappers, ACC API client

src/input/ - Webhook receivers (Fireflies, ACC), Pollers (Gmail, Calendar)

src/output/ - Action gateway, notifications, feedback loop

src/system/ - Data processing, context enrichment, domain processing, decision intelligence

dashboard/ - React frontend (Vite + Tailwind)

dashboard/src/components/ - 15+ React components

config/ - Settings, event type configs (YAML), LLM prompts

tests/ - 235+ unit, integration, and API tests
```

3. AI Pipeline (v1.0)

The core AI pipeline processes construction events through 7 stages, transforming raw communications into actionable coordination proposals. Each stage is validated with Pydantic v2 models and logged to the audit trail for compliance and quality analysis.

Stage	Component	AI Model	Input	Output
1. Capture	Webhooks + Pollers	None	Raw email/transcript/ACC event	RawEvent
2. Normalize	normalizer.py	Haiku 4.5	RawEvent JSON	NormalizedEvent (structured fields)
3. Route	router.py	Haiku 4.5	NormalizedEvent	RoutedEvent + YAML config
4. Enrich	enrichment.py	None (retrieval)	RoutedEvent	EnrichedEvent (ACC + pgvector context)
5. Process	processor.py	None (rules)	EnrichedEvent	ProcessingResult (signals, policies)
6. Propose	proposal_generator.py	Sonnet 4	ProcessingResult	Proposal (actions, confidence)
7. Score	confidence_scorer.py	None (math)	Proposal + history	ConfidenceResult (0-100)

3.1 Confidence Scoring Formula

The confidence scorer uses a 4-factor weighted formula (no LLM, pure math):

Factor	Weight	Calculation
Data Clarity	30%	NormalizedEvent.confidence / 100
Historical Match	25%	Mean cosine similarity of top-3 historical matches
Cost Acceptable	25%	1.0 if cost <= \$50K, 0.0 if > \$50K
No Red Flags	20%	1.0 if no escalation, 0.5 if escalate flag set

Score > 80 = recommend accept. Score 50-80 = recommend review. Score < 50 = recommend reject. PM always makes the final decision (human-in-the-loop enforced).

4. Communication Intelligence (v2.0)

Version 2.0 transforms GigAI from a material-change-only tool into a full communication intelligence platform. 37 requirements across 7 features, all implemented.

Phase	Feature	Key Capabilities	Data Points
10	Unified Inbox	AI classification (5 types), action item extraction, project routing, multi-44	44 messages, 5 sources
11	RFI Automation	Auto-detection from inbox, Sonnet-drafted responses citing knowledge base, 153	153 RFI categories
12	Decision Tracker	Auto-capture from meetings/emails, search, timeline, contradiction detection, 18	18 decisions, tag-based search
13	Stakeholder Map	Relationship graph, notification chains by change type, personalized notifications, 36	36 people, 20 projects
14	Impact Simulator	Earned Value Analysis (SPI/CPI/EAC), Monte Carlo simulation (P50/P75/P90), 1500	1500 budgets breakdown
15	Auto Reports	Daily digest (7 sections), weekly status (9 sections), AI risk identification, 1	Aggregates all data sources
16	Smart Notifications	Urgency scoring (0-100), relevance per recipient, 3-tier routing (immediate, 44	44 notifications, 4 users

5. Data Sources & ACC Integration

GigAI pulls data from 16 ACC API categories plus external sources. All data flows through the facade pattern — real ACC API when credentials are valid, mock fallback for development/demo. Every data point shown in the dashboard comes from (or would come from) these production sources.

ACC API	Data Pulled	Used By
Account Admin	Projects, users, companies, roles	Project library, employee directory, stakeholder graph
Cost Management	Budgets (10 CSI divisions), contracts, change orders, cost impact	Impact simulator budget analysis, project financials, reports
Issues API	Open/closed tasks, quality issues, assignments	Inbox feed, proposal actions, task tracking
RFIs API	Formal RFIs with responses, due dates	RFI queue, auto-drafting, knowledge base
Submittals API	Product data, shop drawings, spec references	RFI drafting, knowledge base context
Documents API	Folder structure, file listings, versions	Drawing markup, document references
Locations API	Floor plans, zones, unit descriptions	Event enrichment, spatial context
Schedule API	Activities, milestones, dates, disciplines	Schedule predictions, impact simulator
Forms/Checklists	Safety inspections, pre-pour checklists	Quality tracking, daily reports
Photos API	Site photos with dates, locations	Progress documentation, reports
Daily Logs	Weather, manpower (by trade), equipment, notes	Reports, resource tracking, field intel
Notifications	In-app ACC notifications to users	Action execution (notify stakeholders)

5.1 External Data Sources

Source	Integration	Data
Fireflies.ai	Webhook + polling (60s)	Meeting transcripts with speaker attribution
Gmail	OAuth polling (60s)	Project emails, RFI responses, supplier quotes
Google Calendar	OAuth polling	Meeting schedules, deadlines
Anthropic Claude	REST API	AI classification (Haiku), proposal generation (Sonnet)
Voyage AI	REST API	1024-dim embeddings for semantic search

6. Prediction Models

All predictions use academically-grounded methodologies, not heuristic guesses. The system implements three prediction frameworks used throughout the construction industry.

6.1 Earned Value Analysis (EVA)

Per PMI PMBOK Chapter 7, EVA calculates project performance using planned value (PV), earned value (EV), and actual cost (AC). Key indices: SPI (Schedule Performance Index) = EV/PV indicates schedule health; CPI (Cost Performance Index) = EV/AC indicates budget health. SPI < 1.0 means behind schedule; CPI < 1.0 means over budget. EAC (Estimate at Completion) = BAC/CPI projects the final cost. When a change is proposed, GigAI recalculates SPI with the delay factored in and projects the new completion date.

6.2 Monte Carlo Simulation

Per AACE International Recommended Practice 44R-08, Monte Carlo simulation models task duration uncertainty using PERT triangular distributions. For each task: optimistic duration = base * 0.8, most likely = base + delay, pessimistic = base * 1.4 + delay. The simulation runs 1,000 iterations, each sampling from these distributions to produce a probabilistic completion date. Output includes P50 (50% probability), P75, and P90 confidence levels, plus a confidence interval. Critical path tasks use higher uncertainty (25% vs 15%) to account for their greater schedule impact.

6.3 Urgency Scoring Model

The schedule urgency scorer evaluates each active task on 5 factors: days until start date (overdue = +40pts), days until finish date (overdue = +40pts), critical path membership (+15pts), actual vs expected progress comparison (behind by >20% = +15pts), and remaining float (0 days = +5pts). Scores are capped at 100. Tasks scoring 80+ are flagged as urgent, 50-79 as at-risk.

6.4 Resource Conflict Detection

The system scans all active tasks for resource double-booking: same crew assigned to tasks with overlapping date ranges. Conflicts are classified as high severity when either task is on the critical path. This prevents scheduling errors that compound into cascading delays.

7. Security & Compliance

All secrets stored in .env file, never hardcoded. ACC API uses 2-legged OAuth2 with cached tokens. Gmail uses OAuth refresh tokens. Human-in-the-loop enforced: nothing executes without PM approval. Every LLM call logged to audit trail (ai_runs table) with token counts and latency. No customer data sent to third parties beyond the configured AI providers (Anthropic, Voyage). SQL injection prevented by parameterized queries throughout. XSS prevented by React's default escaping. CORS configured for dashboard origin only.

8. Deployment & Operations

Backend: uvicorn ASGI server with hot reload. Frontend: Vite dev server + production build (dist/). Database: PostgreSQL 15 with pgvector extension for vector similarity search. The application starts and serves all v2.0 features without PostgreSQL (in-memory stores). When DB is available, proposals, feedback, and knowledge chunks are persisted. Background polling runs every 60 seconds for Fireflies and Gmail. Data seeding happens automatically on startup: 44 inbox items, 17 RFIs, 8 decisions, 30 projects, 30 employees, 24 schedule tasks, 44 notifications.

9. Test Coverage

Test Category	Count	Coverage
Unit Tests (models, store, pipeline)	200+	All Pydantic models, stores, pipeline stages
API Tests (FastAPI routes)	20+	All REST endpoints, error cases
Integration Tests	10+	Full pipeline end-to-end, ACC integration
Frontend Tests (Vitest)	16	Components, hooks, API client
Total	235+	All critical paths

10. API Reference (Summary)

Endpoint Group	Base Path	Methods
Health	/api/health	GET
Proposals	/api/proposals	GET, POST (decision)
Inbox	/api/inbox	GET (list, stats, archived), POST (read, archive)
RFIs	/api/rfis	GET (list, stats), POST (draft, edit, send, close)
Decisions	/api/decisions	GET (list, stats, timeline), POST (check)
Projects	/api/projects	GET (list, stats, detail)
Employees	/api/employees	GET (list, stats, detail)

Schedule	/api/schedule	GET (list, tasks, analysis)
Stakeholders	/api/stakeholders	GET (graph, chain), POST (draft-messages)
Impact Simulator	/api/impact	POST (simulate), GET (EVA)
Reports	/api/reports	GET (daily, weekly)
Notifications	/api/notifications	GET (list, stats, config), POST (config, read)
ACC Data	/api/acc	GET (data, budgets, contracts, etc.)
Learning	/api/learning	GET (stats, history)
SSE Events	/api/events	GET (Server-Sent Events stream)